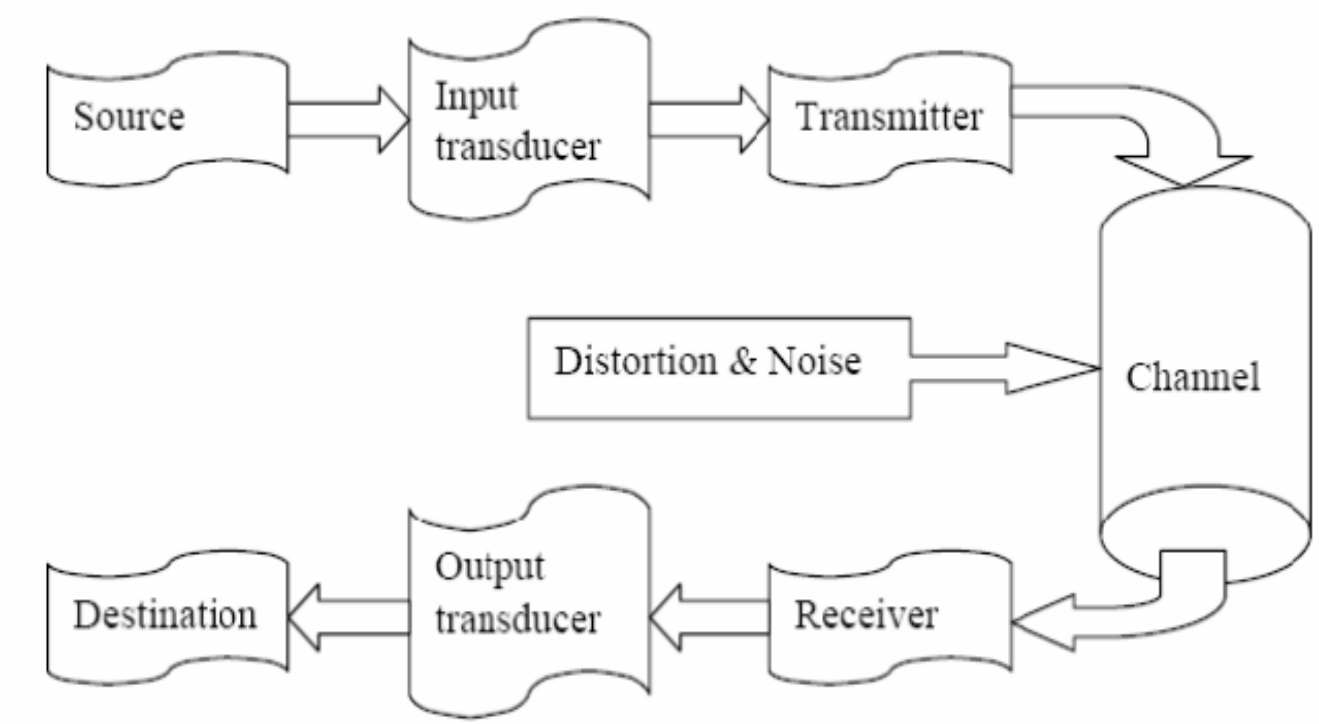
EE3408

ANALOG COMMUNICATIONS

INTRODUCTION

EE3308

Communication System



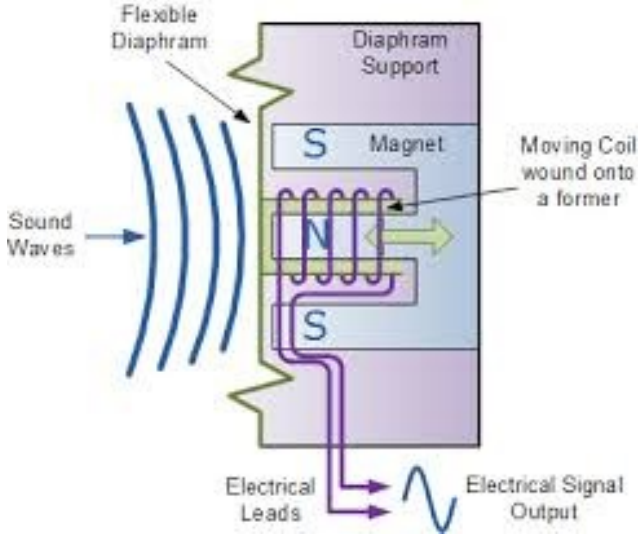
Communication System

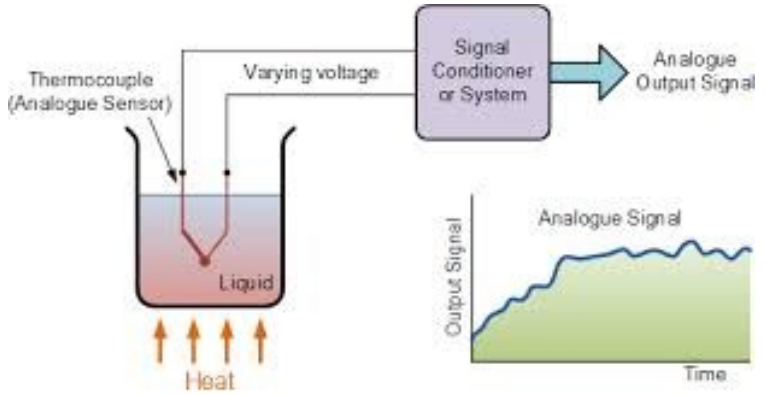
- Source - originates massage





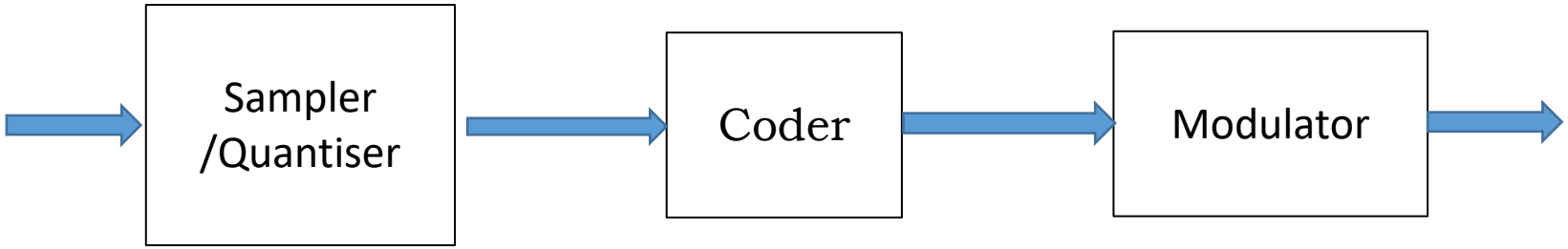
- Input Transducer - converts non electrical data to electrical waveform





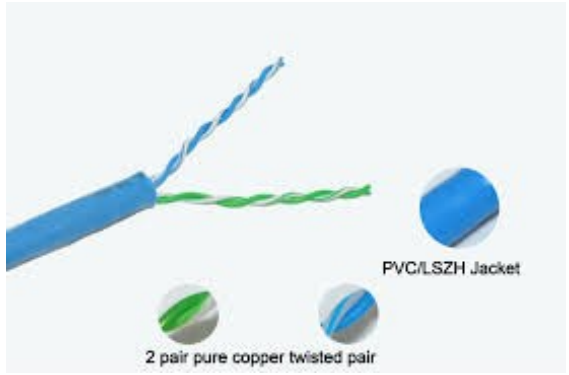
Communication System …/2

- Transmitter - modifies the baseband signal for efficient transmission

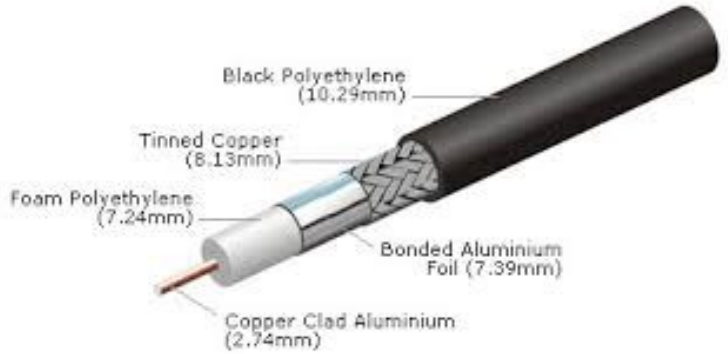


Communication System …/3

- Channel - medium through which the transmitter output is sent



wire



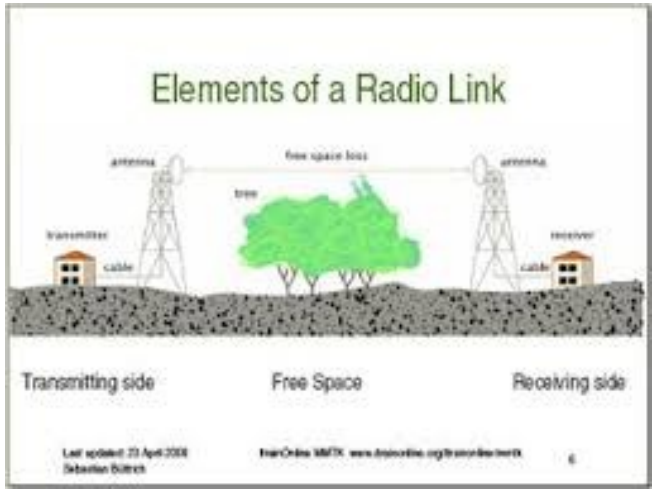
Coaxial cable



waveguide

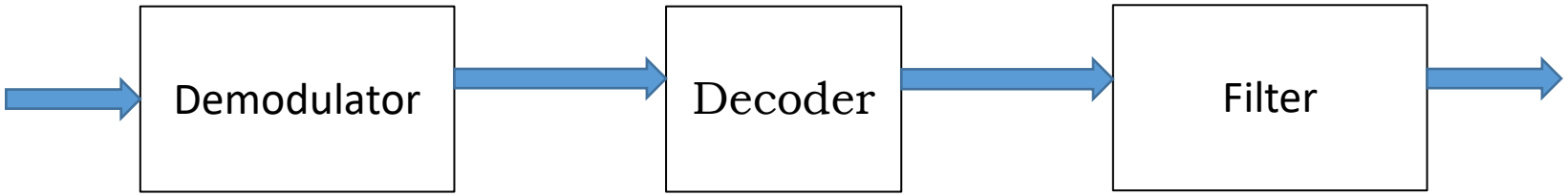


Optical fibre

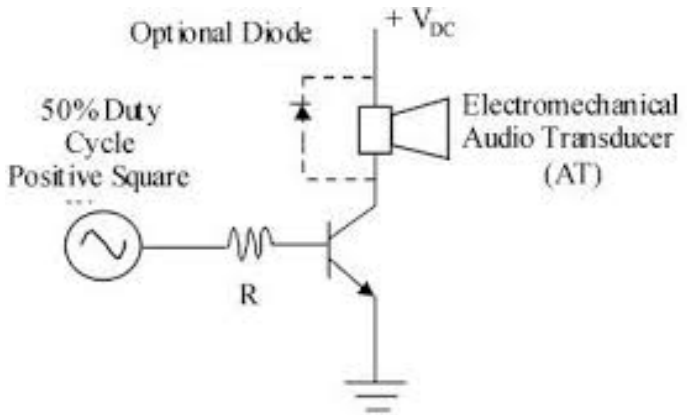


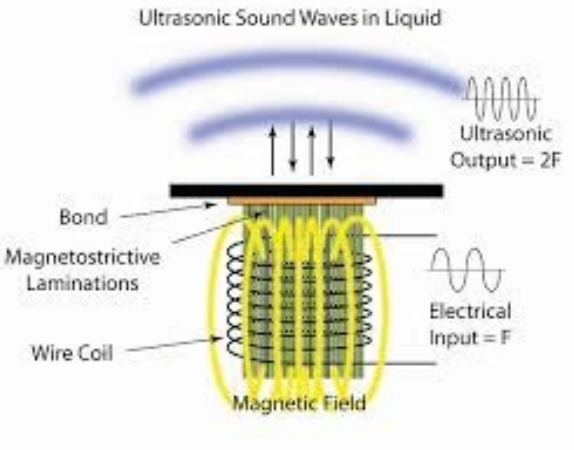
Communication System …/4

- Receiver - reprocesses the signal from the channel



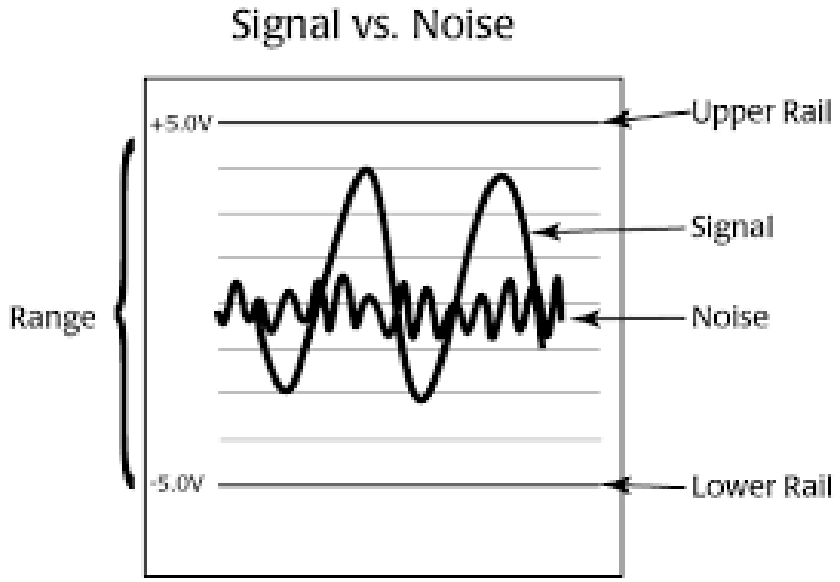
- Output Transducer
  - ➢converts the electrical signal to its original form

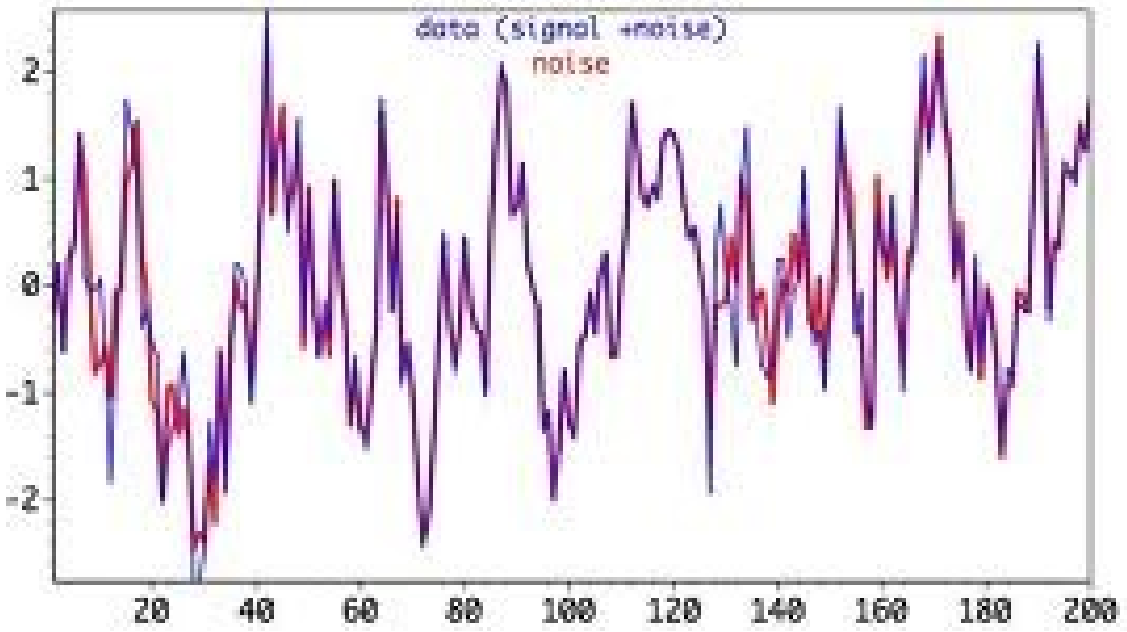




Communication System …/5

- Noise
  - ➢Unwanted signals which can corrupt/interfere with the wanted signal

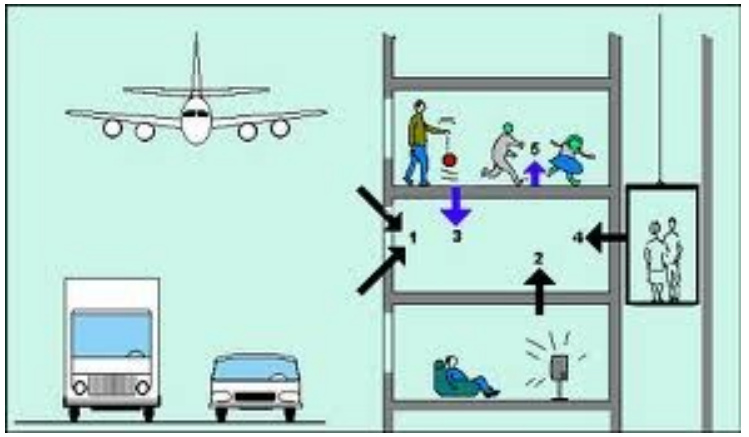




Communication System …/6

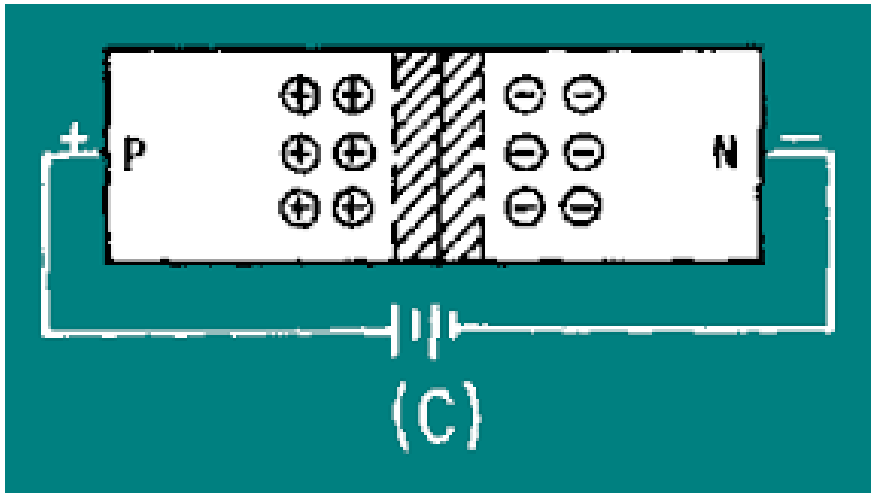
- External noise
  - ➢due to interference from signals transmitted on nearby channels
  - ➢From various sources





Communication System …/7

- Internal noise
  - ❖ Thermal noise
  - ❖ Shot (Semiconductor) noise
  - ❖ Intermodulation (distortion)
  - ❖ Partition noise
  - ❖ Excess noise



Communication System …/8

- Signal to Noise Ratio (SNR)

$$SNR = \frac{Signal\ Power}{Noise\ Power}$$

  - Amplification of received signal to make up for attenuation is of no avail?
  - Amplification further deteriorates SNR?

Need for Modulation

- Baseband signals may not be suitable for direct transmission over a given channel

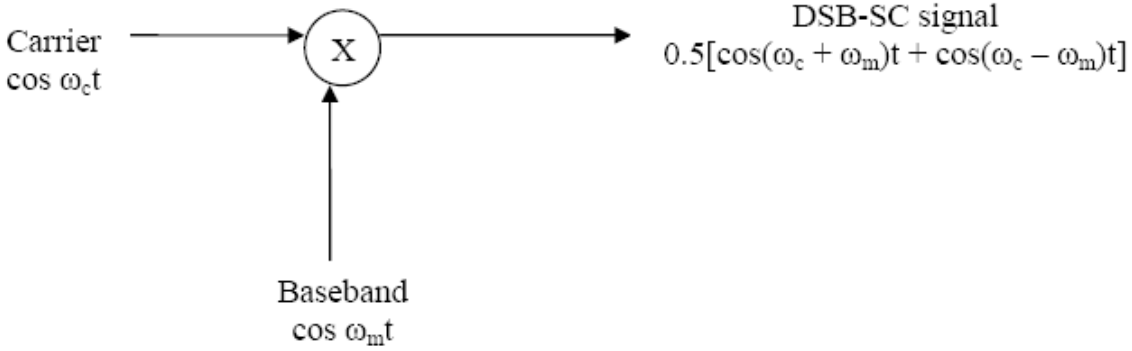
- Modulation -
  - ➢ is used to modify signals to facilitate transmission
  - ➢baseband signal is used to modify some parameter of carrier signal

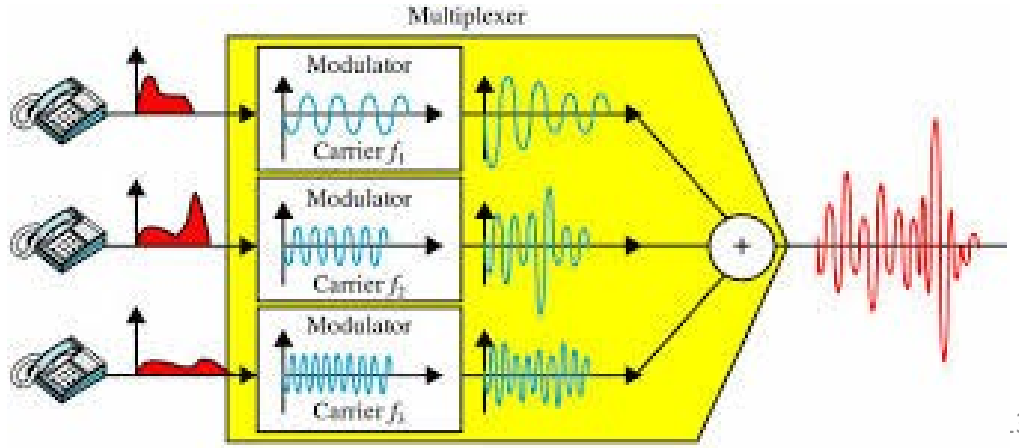
  - ❖Carrier signal - a sinusoid of high frequency
    - ❑ One of its parameters (amplitude, frequency or phase) is varied in proportion to the baseband
    - ❑ AM, FM, PM

Why Modulation is Needed

- Ease of radiation
  - ➢For efficient transmission of electromagnetic energy:
    - ❖ radiating antenna size = one-tenth or more of $\lambda_{radiated\ signal}$

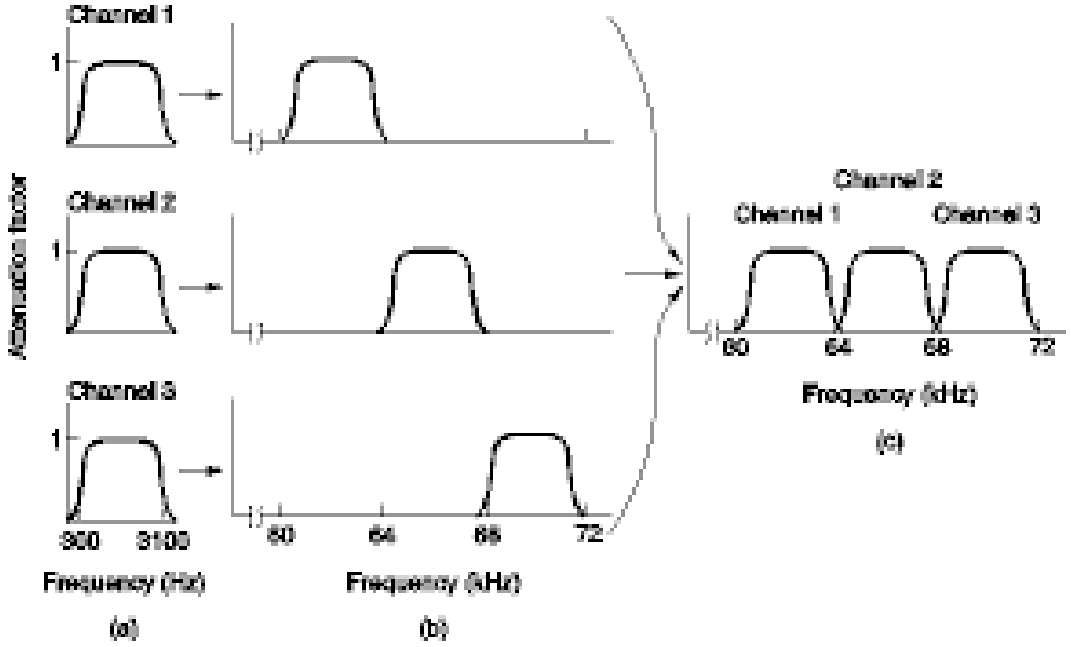
- Simultaneous transmission of signals
  - various signals can be used to modulate different carrier frequencies

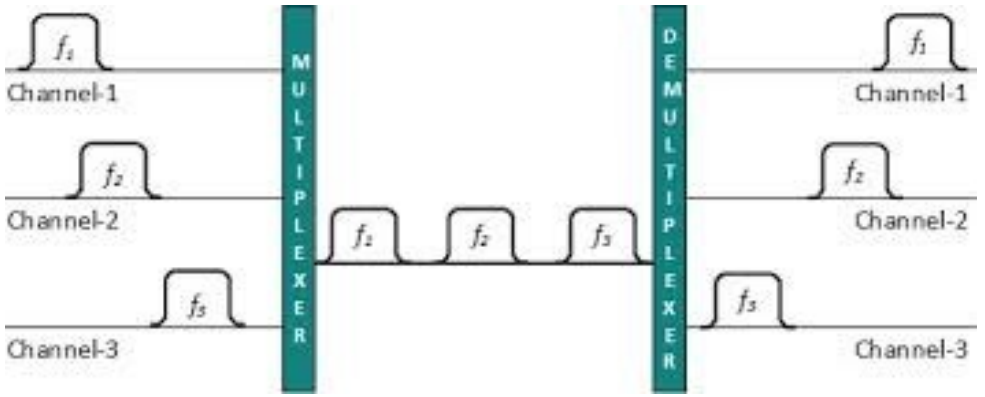




Why Modulation is Needed

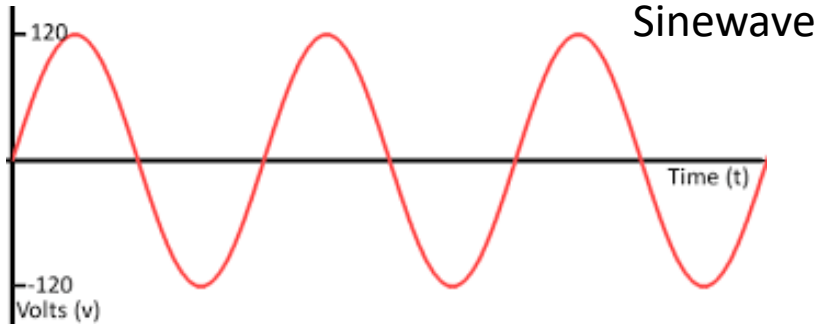
- Simultaneous transmission of signals …….
  - ❖ If the carriers are chosen sufficiently far apart in frequency,
    - ➢ the spectra of the modulated signals will not overlap and thus will not interfere with each other (FDM)

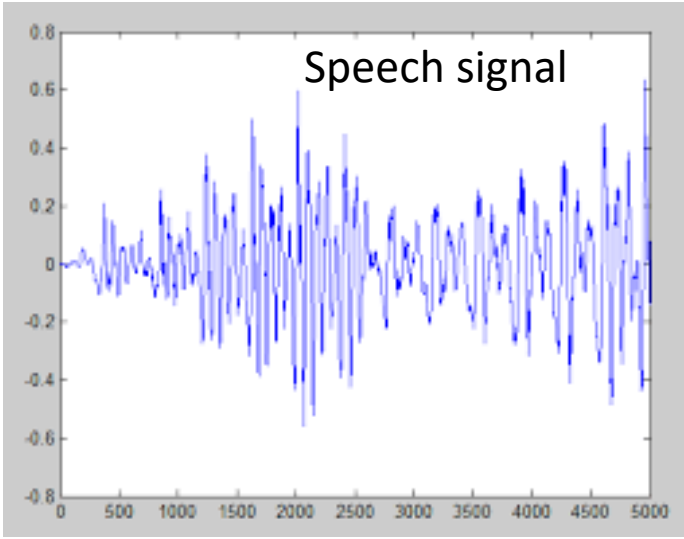




Messages

- Can either be analog or digital
- Analogue messages
  - ➢ Are characterized by data whose value varies over a continuous range

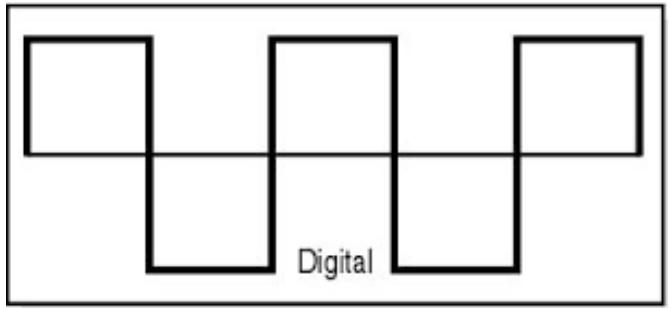




  - ➢vary over a continuous range and can assume infinite possible value

Messages…. /2

- Digital messages
  - ➢ are constructed with a finite number of symbols
  - ➢E.g. Morse coded signal
  - ➢M-ary message
    - ➢ digital message constructed with M symbols



Morse code

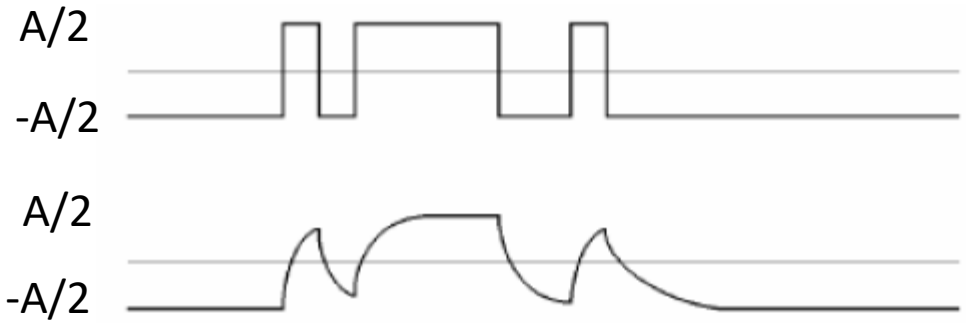
A	.-	N	-.	0	-----
B	-...	O	---	1	.----
C	-.-.	P	.--.	2	..---
D	-..	Q	--.-	3	...--
E	.	R	.-.	4	....-
F	..-.	S	...	5	.....
G	--.	T	-	6	-....
H	....	U	..-	7	--...
I	..	V	...-	8	---..
J	.---	W	.--	9	----.
K	-.-	X	-..-	Full stop	.-.-.-
L	.-..	Y	-.--	Comma	--..--
M	--	Z	--..	Query	..--..

Messages…. /3

- Digital messages
  - Morse coded message
    - Electrical pulse A/2 = mark; -A/2 = space
  - M-ary case
    - M distinct electrical pulses (or waveforms) are used;
    - each of the M pulses represents one of the M possible symbols
    - $M = 2^k$, k = number of bits
  - E.g. 4-Ary case - (k = 2 bits)
    - Will have 4 distinct waveforms

Messages…. /4

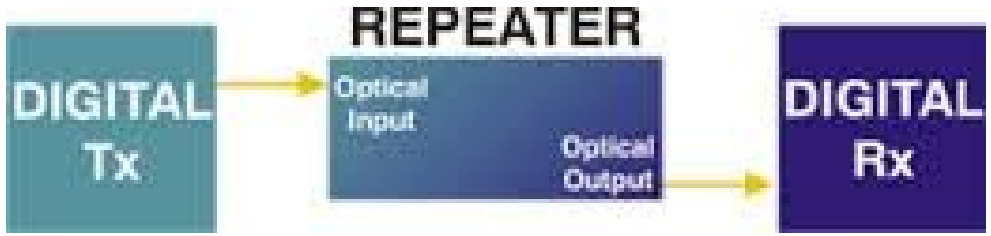
- Task of Receiver
  - ➢ extract message from distorted & noisy signal at channel output



  - ➢ Message extraction is easier from digital than from analog signals
  - ➢Decision is made with reasonable certainty even if pulses are distorted & noisy

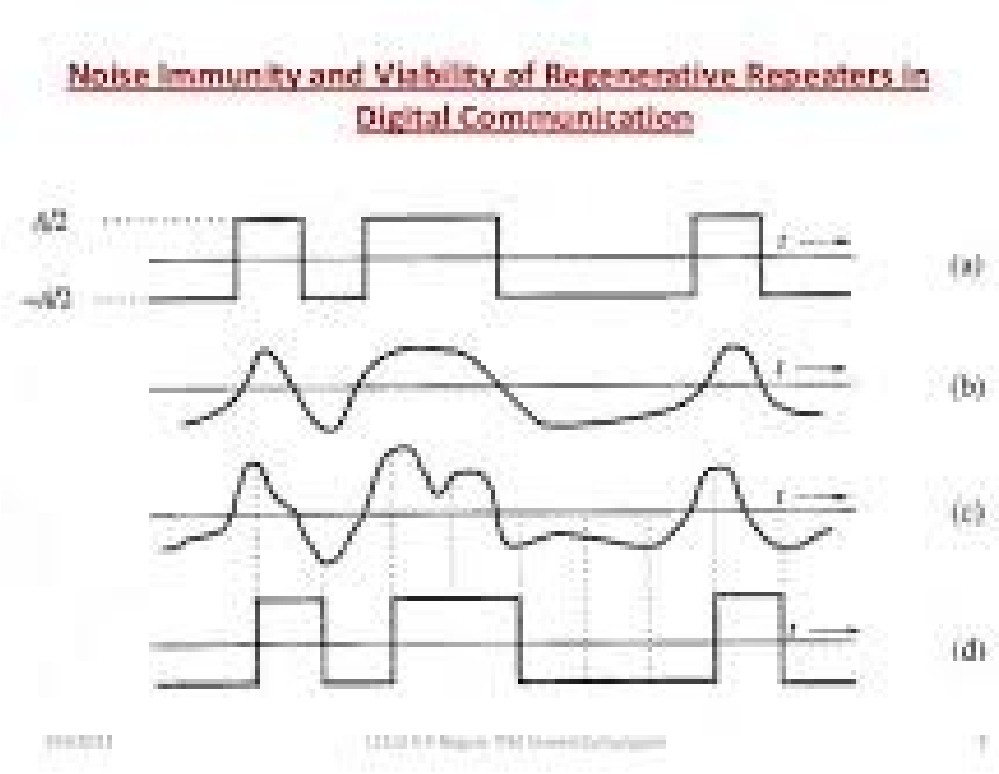
Messages…. /5

- In the presence of noise & distortion
  - ➢Digital comm. System can transmit messages with greater accuracy than analog system
- Regenerative Repeaters can be used for digital comms



Messages…. /6

- Repeater station detects pulses and transmits new clean pulses
  - ➢combating the accumulation of distortion and noise



Messages…. /7

- waveform in analog messages is important …. ?
- A regenerative repeater is not viable for analog signals …?
- Distance over which an analog message can be transmitted is limited by transmitter power

Decibels

- One tenth of a Bell
- used all electronic communications to represent relative parameters
  - gain or loss and absolute parameters such as signal amplitude in volts and watts

$$G(dB) = \ 10\ log\frac{P_1}{P_2}$$

  - *P*1 and *P*2 (power levels in Watts)
- Since $P_1 = (V_1)^2/R_1$ and $P_2 = (V_2)^2/R_2$

$$\mathrm{G}(dB) = \ 20\ log_{10}\frac{V_1}{V_2} + \ 10\ log_{10}\frac{R_2}{R_1}$$

  - *V*1 & *V*2 (voltate in Volts) and *R*1 & *R*2 (resistances in Ohms)

Decibels…/2

- Converting signal level in watts to decibels (using 1W as reference value)

$$P(dBW) = \ 10\ log\frac{P}{1W}$$

  - e.g. P = 1000W, P(dBW) = +30 dBW
  - P = $1x10^{-17}\ W$ P(dBW) = -170 dBW

- The dBm system uses 1mW as a reference

$$P(dBm) = \ 10\ log_{10}\frac{P}{1mW} = \ 10\ log_{10}\frac{P}{1x10^{-3}\ W}$$

$$P(dBm) = \ 10\ log_{10}P + 10log_{10}10^{3} = \ P(dbW) + 30$$

Noise

- Originates both in the channel and in the communication equipment
- Consists of undesired, random variations that interfere with the desired signals and inhibit communication
- It cannot be avoided completely, but its effects can be reduced
- Two types of Noise
  - Internal
  - External

Noise ../2

- External Noise
  - ❖If the channel is a radio link, there are many possible sources of noise
    - ➢Equipment noise
    - ➢Atmospheric noise
    - ➢Space noise

Noise ../3

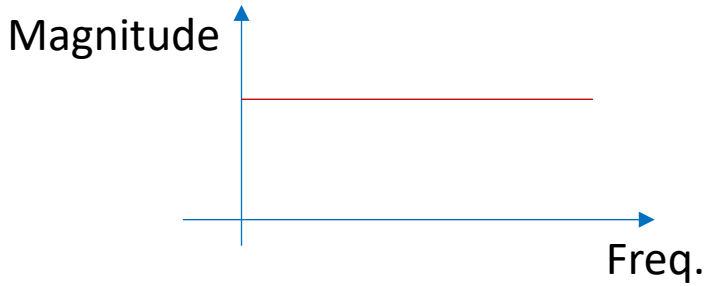
- Internal Noise

❖Generated in all electronic equipment (passive & active devices)

➢Thermal noise

❑is a mixture of noise of all frequencies (white noise)

❑ Noise Power from conductor



$$P_N = kTB$$

k = 1.38*x*10^{-23} (J/K), T = absolute temperature ($K = 273^0 + Celcius\ temp$),
B = bandwidth (noise power)

Noise ../4

- Thermal Noise
  - ❖ Reduction of thermal noise ($P\ \alpha\ TB$)
  - ❖ At any instant in time,
    - ➢ more electrons are moving in some directions than in others
    - ➢ movement of each electron constitutes a very small current
  - ❖ Root mean square (rms) noise voltage

$$V_N = \sqrt{4kTBR}$$

Noise ../5

- Power in resistive circuit
- Assuming $R_L = R_N$

$$V_{R_L} = \frac{V_N}{2}$$

- The rms voltage across the load is

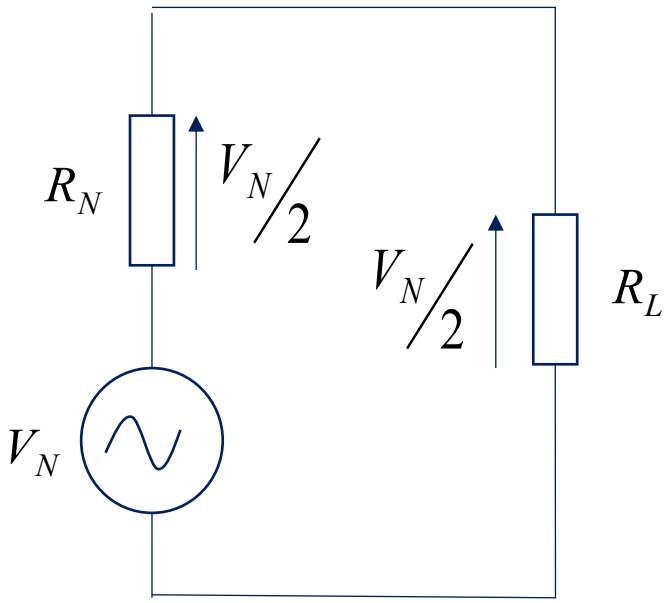
$$V_L = \sqrt{PR}$$

- Since $P_N = KTB$
- Then

$$V_{R_L} = \sqrt{KTBR}$$

- So

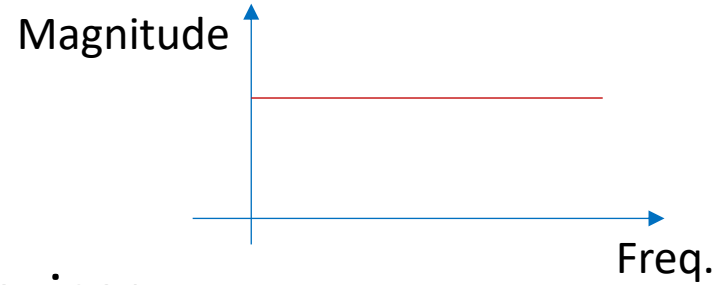
$$V_N = 2V_{R_L} = \sqrt{4kTBR}$$



Noise Resistor

Noise ../5

- Short Noise
  - ❖ Spectrum same as thermal noise



  - ❖Due to random variations in current flow in active devices

  - ❖ RMS short noise current

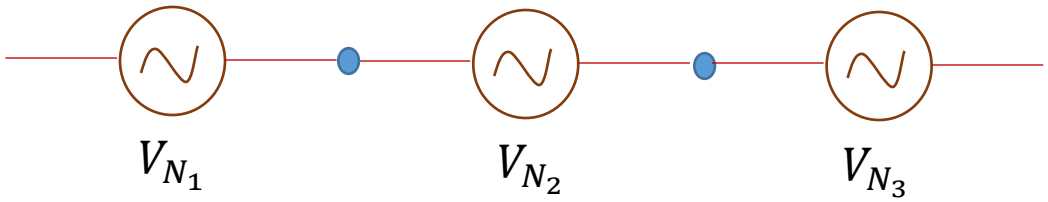
$$I_N = \sqrt{2qI_0B}$$

q = electronic charge = $1.6x10^{-19}$ *C;*

I_0 = dc current flowing across the junction (A); B = Bandwidth

Noise ../6

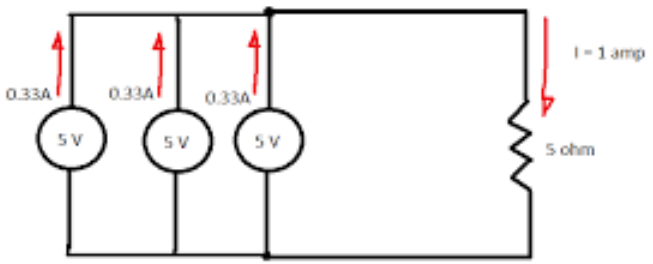
- For voltage sources in series



$$V_{N_t} = \sqrt{V_{N_1}^2 + V_{N_2}^2 + V_{N_3}^2 + \ldots}$$

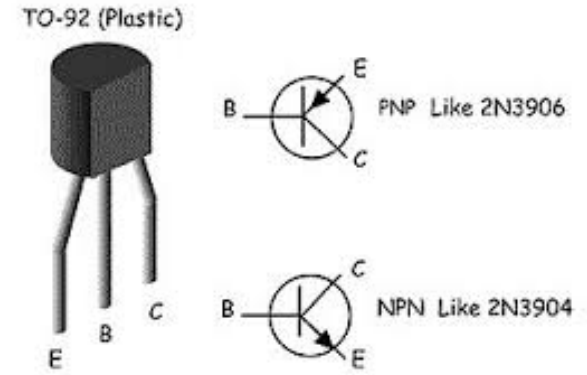
- For current sources in parallel

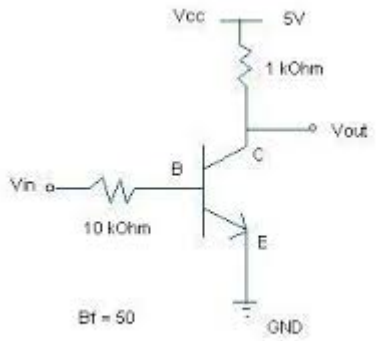
$$I_{N_t} = \sqrt{I_{N_1}^2 + I_{N_2}^2 + I_{N_3}^2 + \ldots}$$



Partition Noise

- Similar to short noise in its spectrum & generation mechanism
- Only occurs in devices where single current separates into two or more paths e.g. BJT
- As charge carriers divide into one stream or the other
  - ❖A random element in the currents is produced



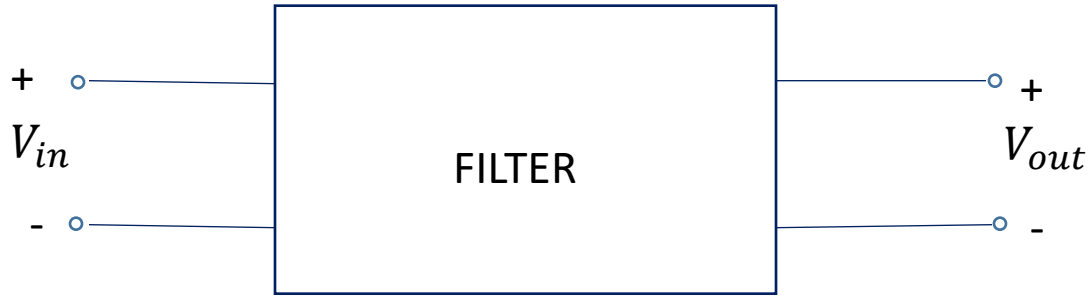


Excess (Flicker or Pink) Noise

- Called flicker noise or 1/f noise or pink noise?
- Is found in tubes but Is problematic in semiconductors and carbon resistors
- Is rarely a problem in communications

Filters

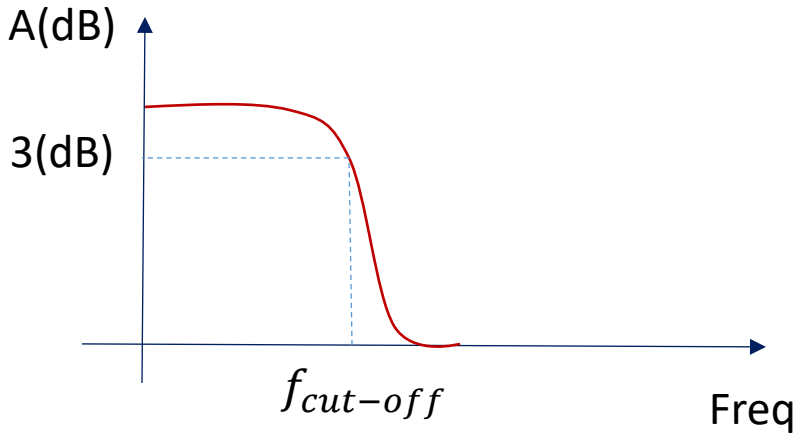
- An electrical network that
  - ❖alters the amplitude and/or phase characteristics of a signal w.r.t frequency



- Are defined by their frequency domain effects on signals
- Plot of filter's voltage gain (dB) vs frequency = Frequency response of filter
- Gain of filter

$$A(dB) = 10\log\frac{P_{out}}{P_{in}} = 20\log\frac{V_{out}}{V_{in}}$$

Filters ../2



- Cut-off frequency
  - Freq. at which Power gain drops to 0.5 or 3dB of maximum value
  - Freq. at which Voltage gain drops to $1/\sqrt{2}$ or 0.707 of maximum value

$$P = V^2$$

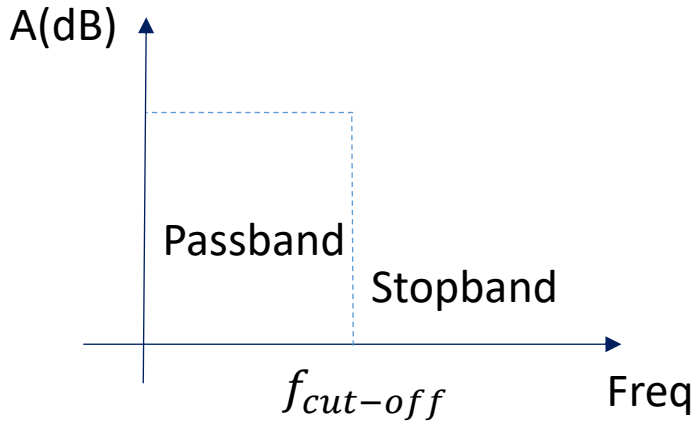
$$\frac{1}{2} = \left(\frac{1}{\sqrt{2}}\right)^2$$

$$P \downarrow \frac{1}{2} \; or \; 3\,dB$$

$$V \downarrow \frac{1}{\sqrt{2}} = 0.707$$

Filters ../3

- Bode Plot (ideal plot of frequency response in the logarithmic scale)



- Filter can be described in terms of its Transfer Function (T.F.)

$$Transfer\ Function\ H(s) = \frac{Output\ Quantity}{Input\ Quantity} = \frac{LT\ of\ V_{out}}{LT\ of\ V_{in}} = \frac{V_{out}(s)}{V_{in}(s)}$$

Filters ../4

- Laplace Transform (LT) of function *f(t)*

$$L[f(t)](s) = \int_0^\infty f(t)e^{-st}dt$$

f(t)	1	t	t^n	e^{at}	$cos\,(\omega t)$
L[f(t)] (s)	$1/s$	$1/s^2$	$\frac{n!}{s^{n+1}}$	$\frac{1}{s-a}$	$\frac{s}{s^2+\omega^2}$

$$f(t) = 1; \quad L[f(t)](s) = \int_0^\infty f(t)e^{-st}dt = \int_0^\infty 1\, e^{-st}dt$$

$$= \left[-\frac{1}{s}e^{-st}\right]_0^\infty = -\frac{1}{s}[0-1] = \frac{1}{s}$$

Filters ../5

- Parameter “s” is complex number

$$s = \alpha + j\omega = j\omega, \quad for\ \alpha = 0$$

- Consider sinusoid

$$V = Ae^{j\omega t} = Ae^{st}; \qquad s = j\omega$$

- For Resistor

$$R = V/I = Z$$

- For capacitor

$$I = C\frac{dV}{dt} = CAj\omega e^{j\omega t} = j\omega CV = sCV; \qquad Z = \frac{V}{I} = \frac{1}{sC} = \frac{1}{j\omega C}; \qquad |Z| = \frac{1}{\omega C}$$

- For inductor

$$I = \frac{1}{L}\int Vdt = \frac{1}{L}\int Ae^{j\omega t}dt = \frac{1}{j\omega L}Ae^{j\omega t} = \frac{1}{j\omega L}V = \frac{V}{sL}; \qquad Z = \frac{V}{I} = sL = j\omega L; \qquad |Z| = \omega L$$

Filters ../6

- In phasors

$$If \quad V_{out} = A + j\omega_1$$
$$V_{in} = B + j\omega_2$$

$$Then \quad |V_{out}| = \sqrt{A^2 + \omega_1^2}, \quad \angle V_{out} = \tan^{-1}\left(\omega_1 / A\right)$$
$$|V_{in}| = \sqrt{B^2 + \omega_2^2}, \quad \angle V_{in} = \tan^{-1}\left(\omega_2 / B\right)$$

$$V_{out} = |V_{out}|\angle\theta$$
$$V_{in} = |V_{in}|\angle\theta$$

- Magnitude of T.F.

$$|H(s)| = \frac{|V_{out}(s)|}{|V_{in}(s)|} = \frac{Z(s)}{P(s)}$$

- Phase of T.F.

$$\theta = \theta_{out} - \theta_{in}$$

Filters ../7

- Roots of Z(s) are called zeros
- Roots of P(s) are poles

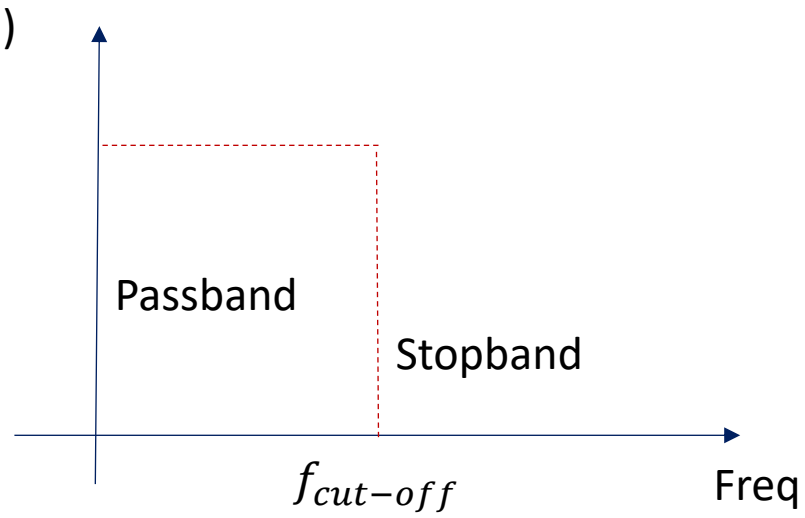
- Order of a filter
  - Is the highest power of the “s” in the T.F.
  - Is given by total No. of inductors & capacitors

  - The higher the filter order, the better in can discriminate between signals at different frequencies

Filters ../8

- Can be classified according to functions they perform on range of frequencies (Passband, stopband)

- PASSBAND
  - ➢ Range of frequencies passed by a filter



- STOPBAND
  - ➢Range of frequencies over which unwanted signals are attenuated

Filters ../9

- Attenuation

$$\alpha = -20\log|H| \;\; dB \qquad |H| \le 1; \qquad |H| = 10^{-\alpha/20} = 10^{-0.05\alpha}$$

- Gain

$$A = \;\; 20\log|H| \;\; dB \qquad |H| > 1; \qquad |H| = 10^{A/20} = 10^{0.05A}$$

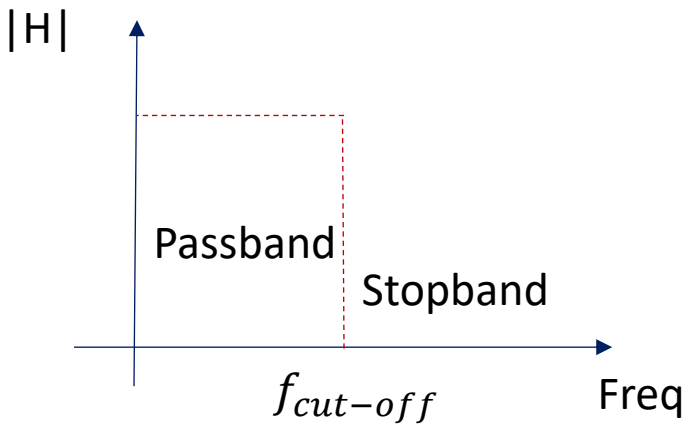
Filters ../10

- In ideal case
  - Passband is such that

$$\alpha = 0 \qquad |H| = 1;$$

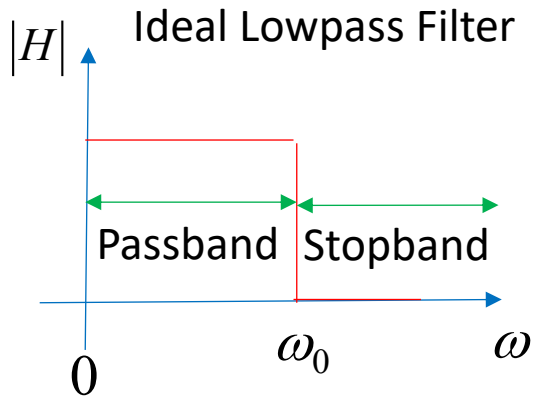
  - Stopband is such that

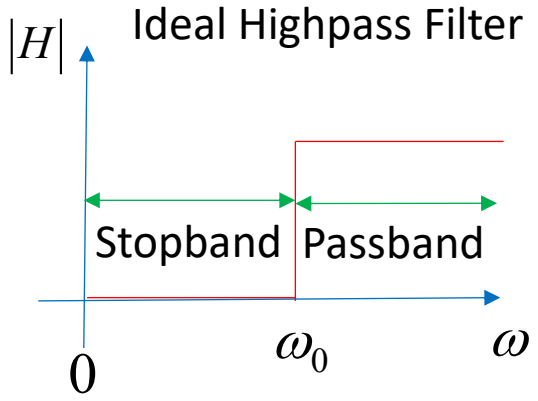
$$\alpha = \infty \qquad |H| = 0;$$

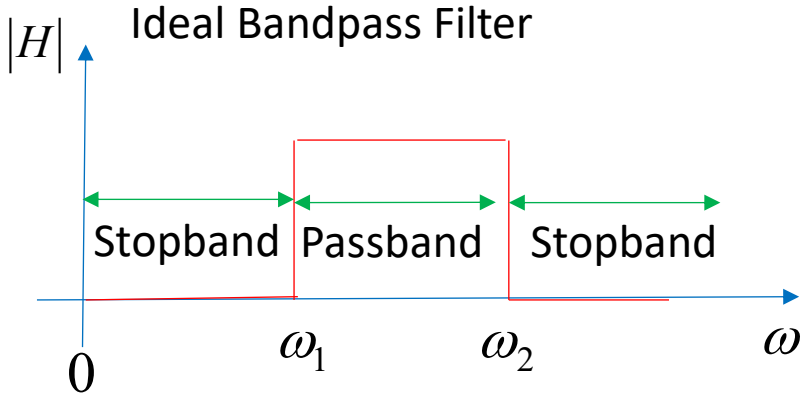


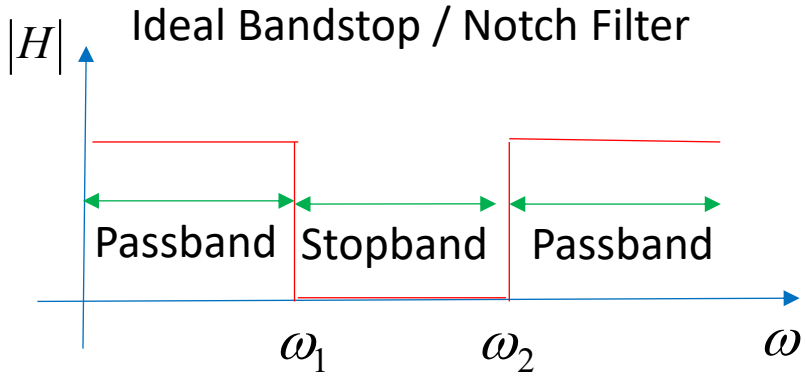
Filters ../11

- Common Filters



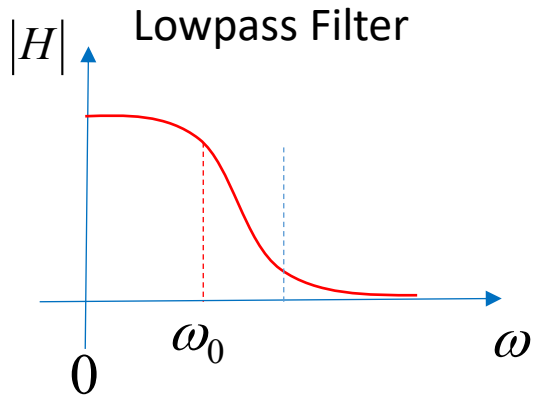


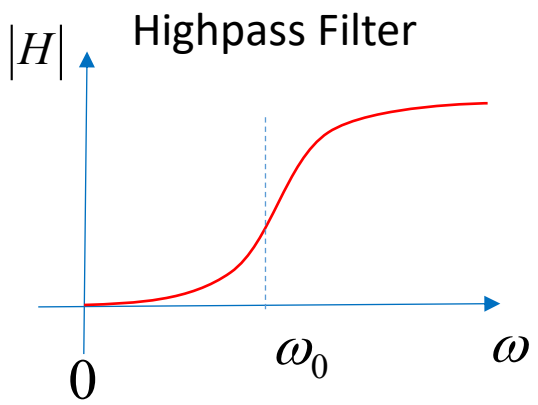


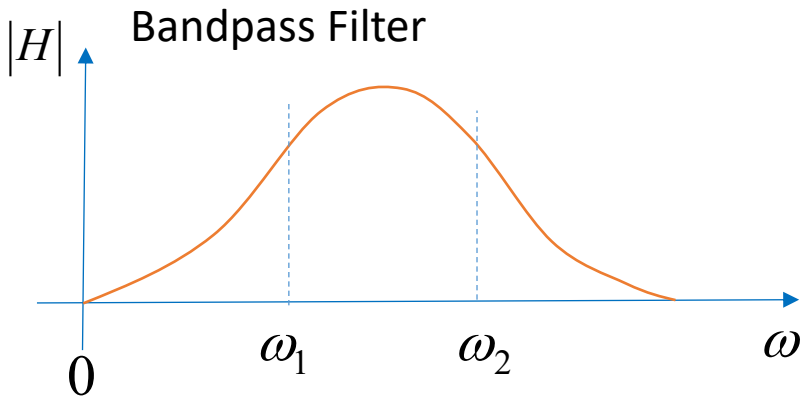


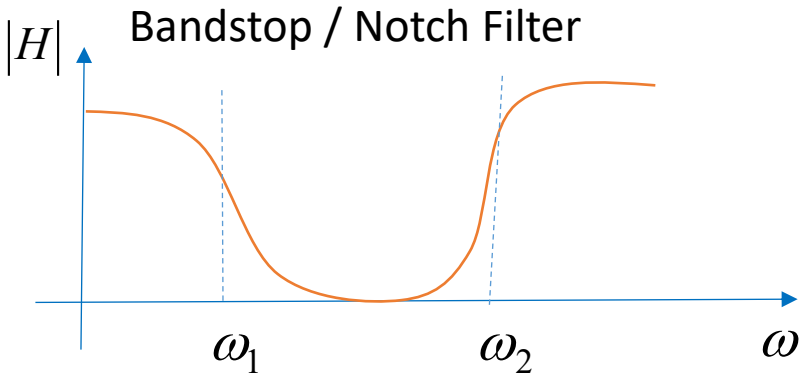
Filters ../12

- Realistic Filters



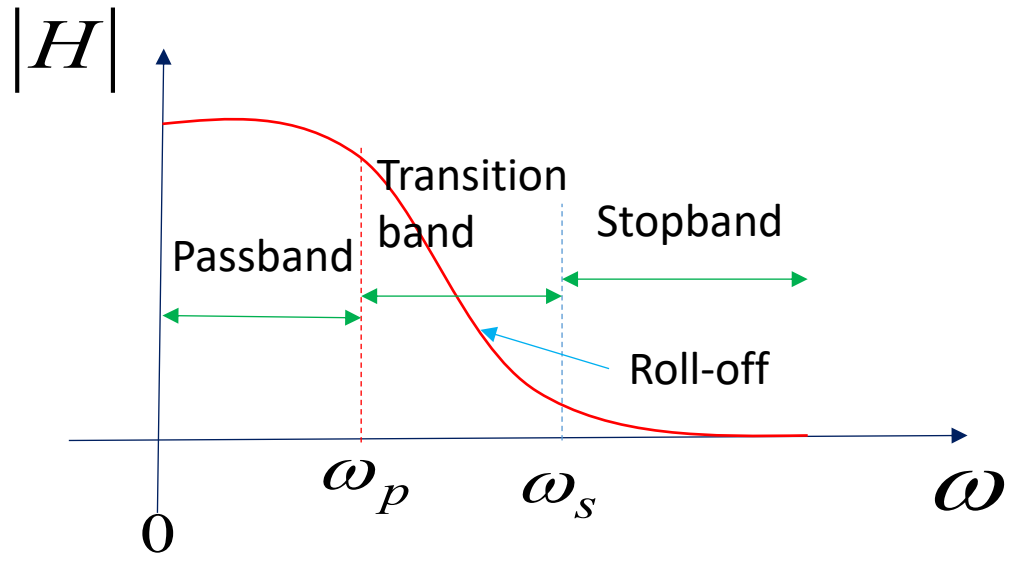






Filters ../13

- The rate of change of attenuation between the passband and stopband is the ROLLOFF



- The slope of the curve in the transition band depends on order of filter
  - Higher order filter has steeper slopes (roll-off)

Filters ../14

- Attenuation slope (α) is usually expressed in
  a) dB/octave (dB per factor of 2)
  b) dB/decade (dB per factor of 10)

- The slope of the curve in the transition band depends on order of filter
  - ➢Higher order filter has steeper slopes (roll-off)

- From Bode plot (dB voltage gain vs freq. plot)
  - ➢ROLL-OFF (asymptotic slope) is measured in multiples of
    - ❖6 dB per octave or
    - ❖20 dB per decade

Filters ../15

Example

$$Say \quad A_2 = 20\log\left(\frac{\omega_2}{\omega_1}\right)$$

- For zeros

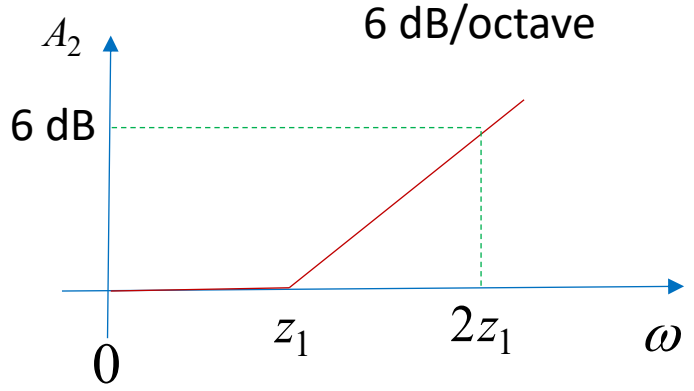
$$If \quad \omega_1 = z_1; \quad and \quad \omega_2 = 2z_1$$

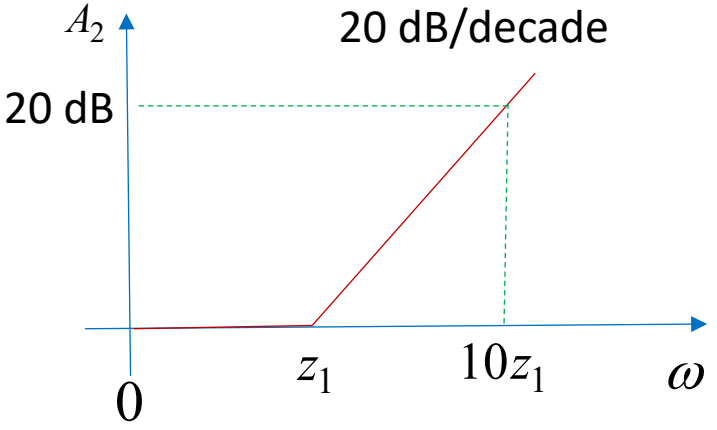
$$A_2 = 20\log\left(\frac{2z_1}{z_1}\right) = 6dB/octave$$

or

$$If \quad \omega_1 = z_1; \quad and \quad \omega_2 = 10z_1$$

$$A_2 = 20\log\left(\frac{10z_1}{z_1}\right) = 20dB/decade$$





Filters ../15

Example

$$Say \;\; \alpha = -20\log\left(\frac{\omega_2}{\omega_1}\right)$$

- For Poles

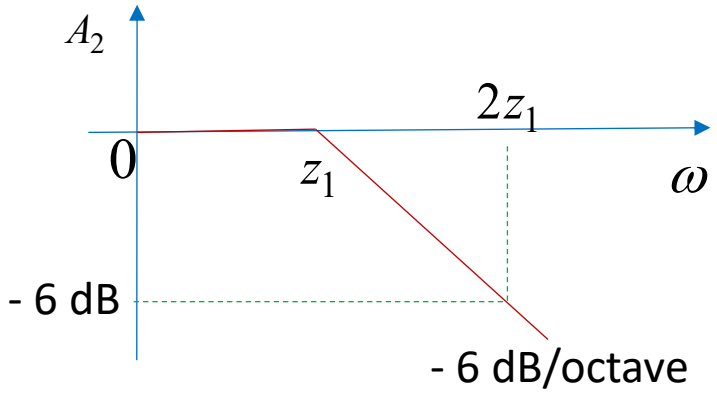
$$If \;\; \omega_1 = z_1; \;\; and \;\; \omega_2 = 2z_1$$

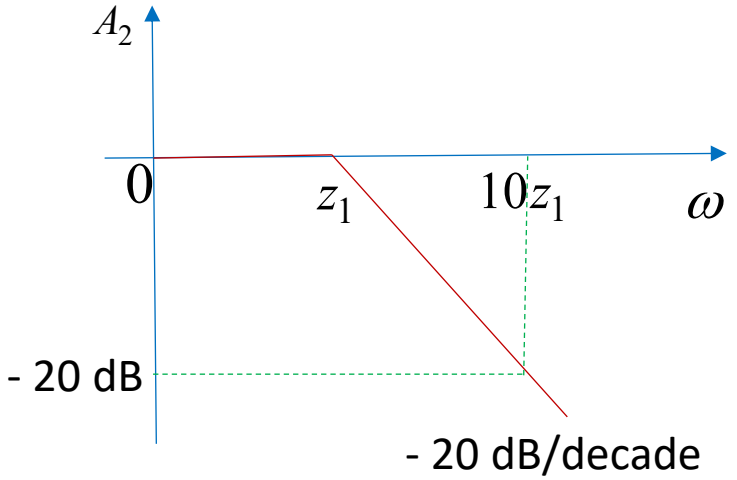
$$\alpha = -20\log\left(\frac{2z_1}{z_1}\right) = -6\,dB/octave$$

or

$$If \;\; \omega_1 = z_1; \;\; and \;\; \omega_2 = 10z_1$$

$$\alpha = -20\log\left(\frac{10z_1}{z_1}\right) = -20\,dB/decade$$





Passive Filters

- Are made of passive components
- See notes

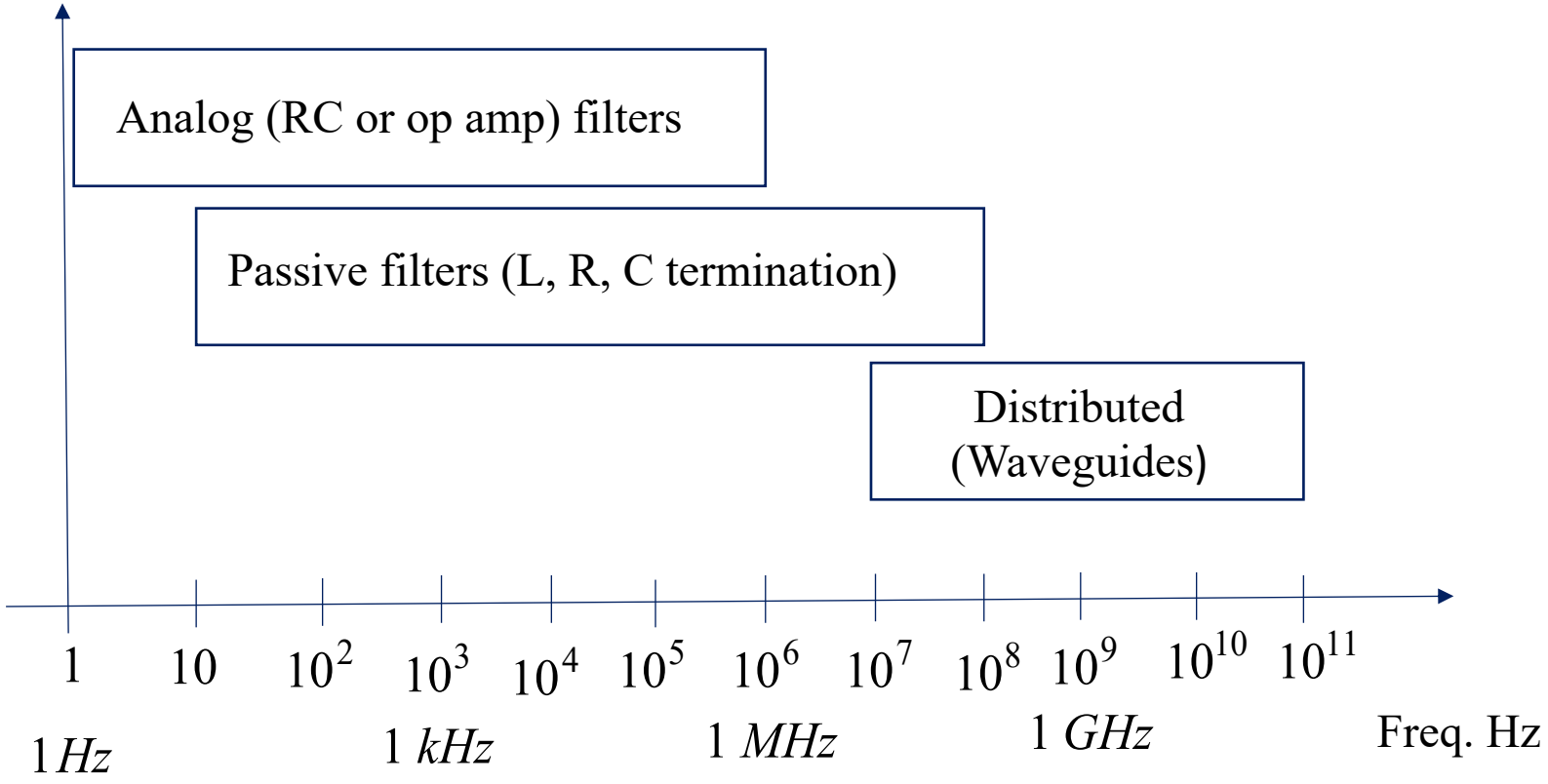
- Disadvantages
  - ➢See Notes

Filters ../17

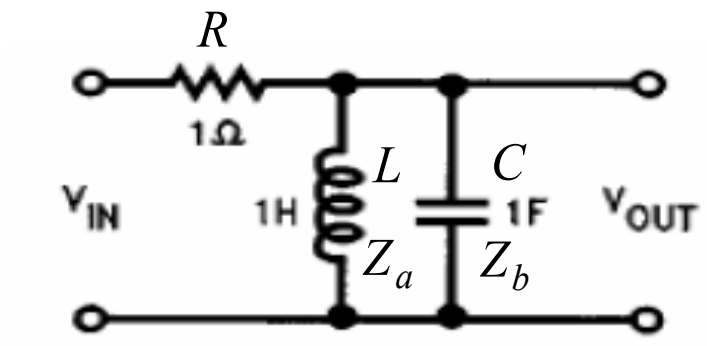
- Factors that influence choice of the filter
  a) Range of frequency of operation
  b) Sensitivity to parameter change and stability
  c) Weight and size of the realization
  d) Availability of voltage sources for the operational amplifier

Filters ../18

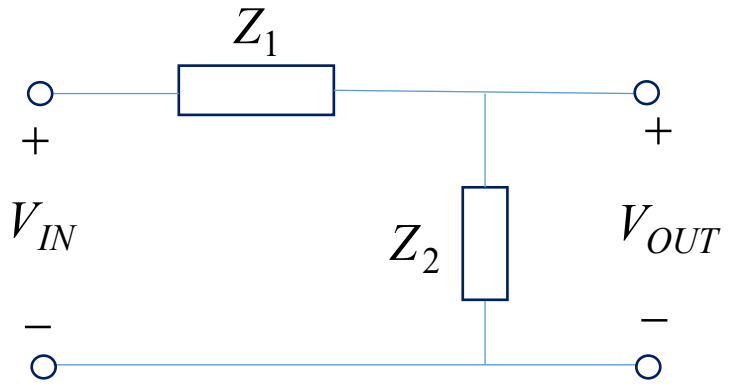
- Frequency of operation of three kinds of filters



Band Pass Filter



$\Rightarrow$



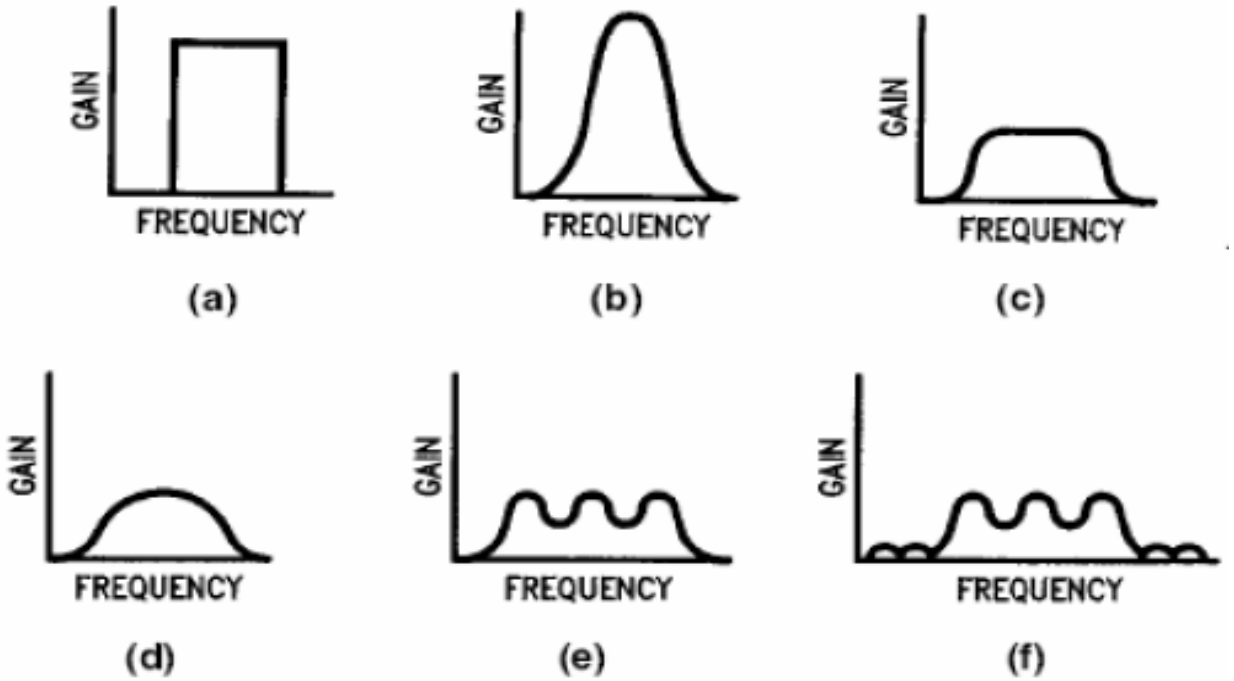
$$H(s) = \frac{V_{OUT}}{V_{IN}} = \frac{Z_2}{Z_1 + Z_2} = \frac{sL/(s^2LC+1)}{R + sL/(s^2LC+1)} = \frac{Ls}{RLCs^2 + Ls + R} = \frac{\frac{1}{RC}s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$For\ R = 1\Omega,\ L = 1H,\ C = 1F$$

$$H(s) = \frac{s}{s^2 + s + 1}$$

Band Pass Filter ../3

- Examples of bandpass filter amplitude response

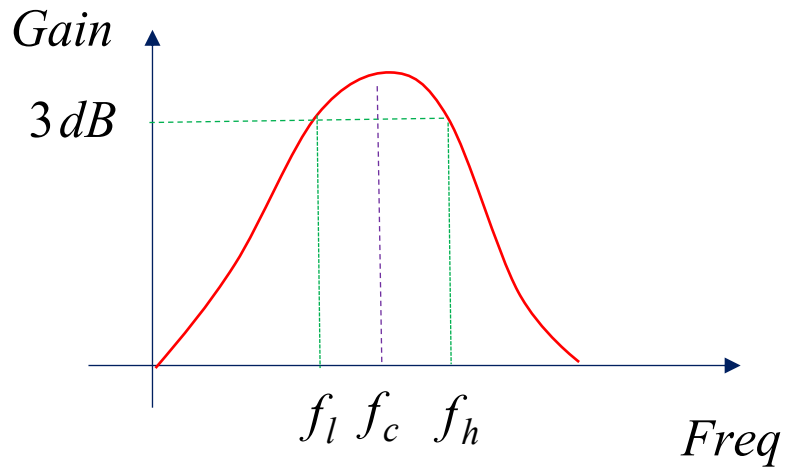


Band Pass Filter ../4

- Bandpass filter amplitude response has peak value at filter's centre frequency (f_c)

- f_c = geometric mean of - 3dB freqs (f_l & f_h)

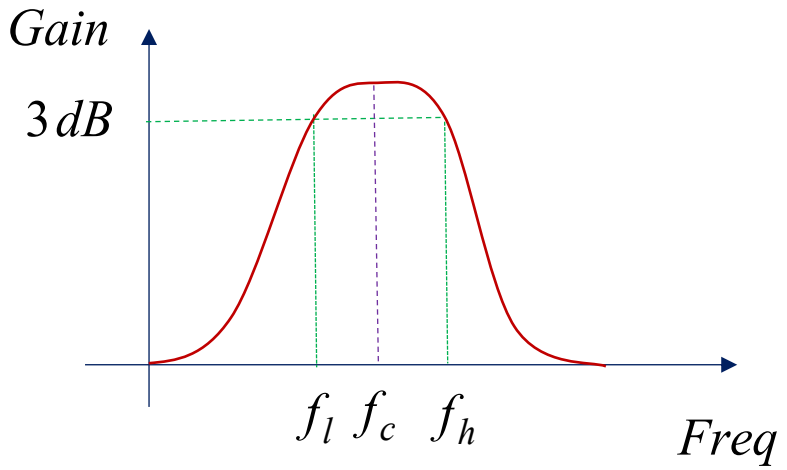
$$f_c = \sqrt{f_l f_h}$$



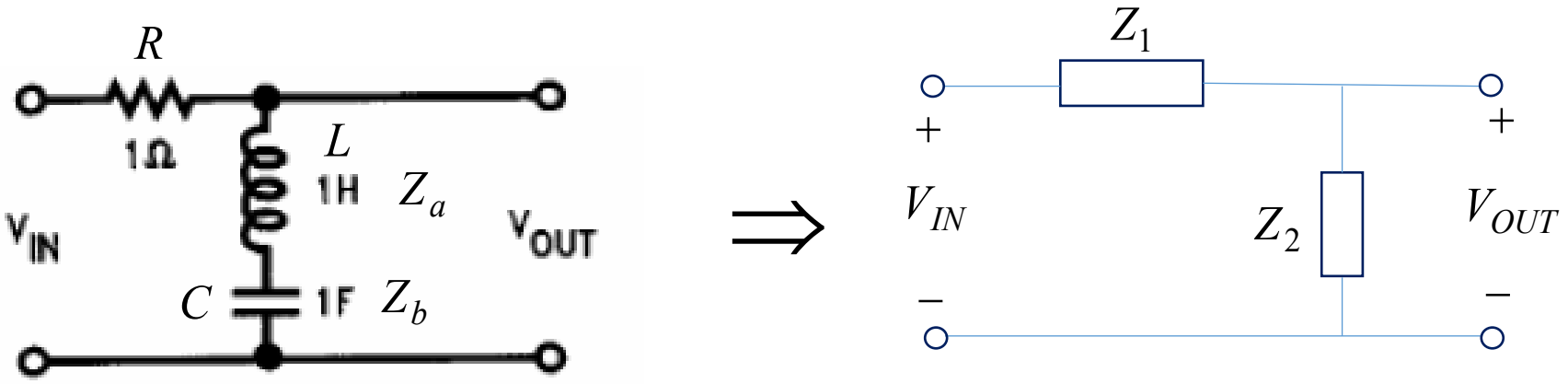
Band Pass Filter ../5

- Quality factor Q
  - ➢is measure of the sharpness of the amplitude response

$$Q = \frac{f_c}{f_h - f_l}$$



Notch or Band-stop Filter



- Transfer function

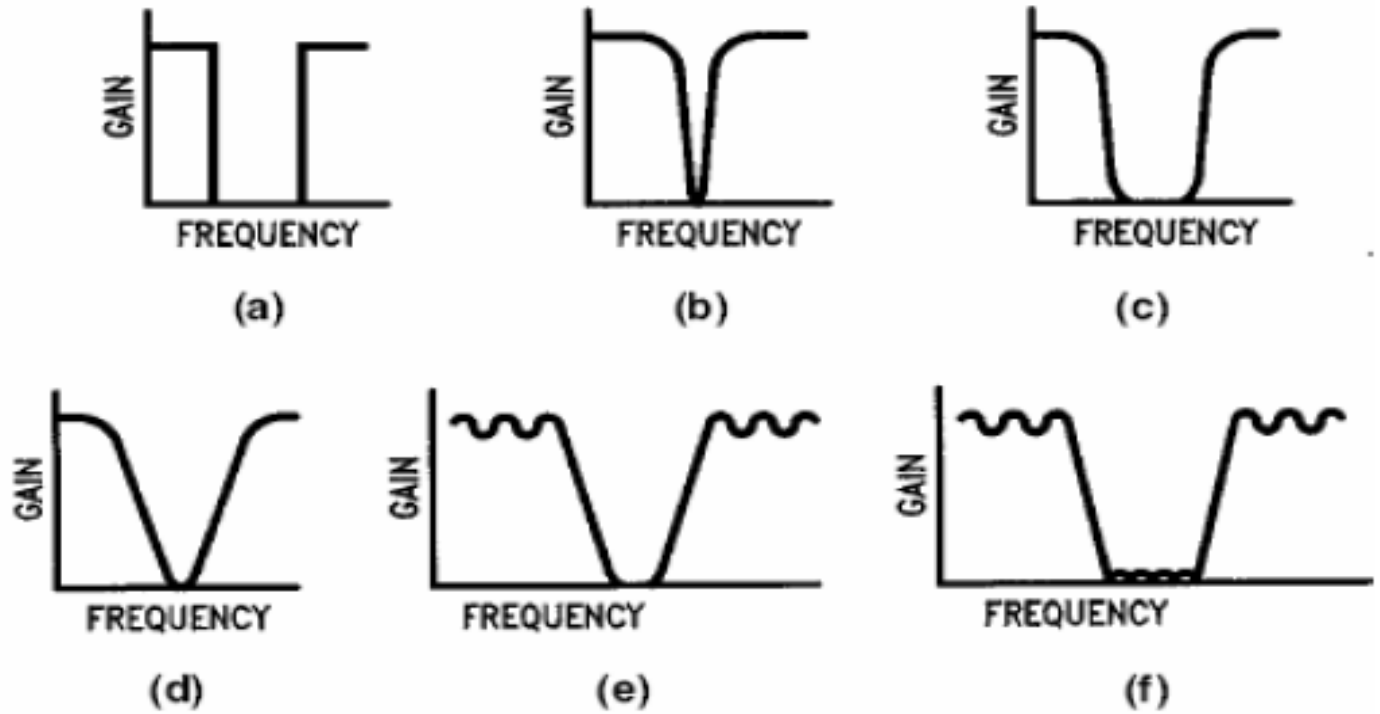
$$H(s) = \frac{V_{OUT}}{V_{IN}} = \frac{s^2 + \frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

$$For\ R = 1\Omega,\ L = 1H;\ C = 1F$$

$$H(s) = \frac{s^2 + 1}{s^2 + s + 1}$$

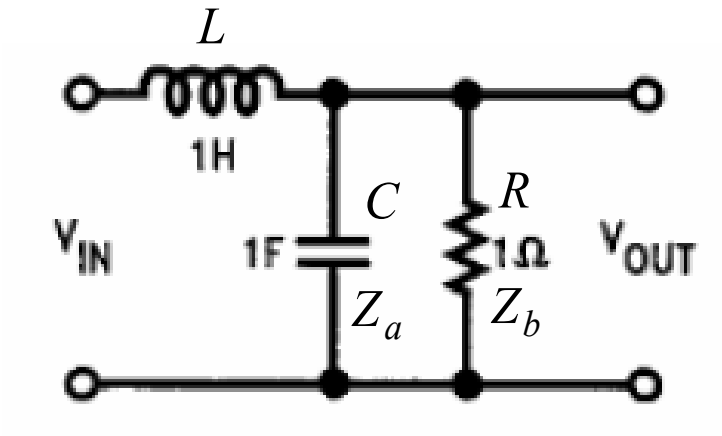
Notch or Band-stop Filter ../3

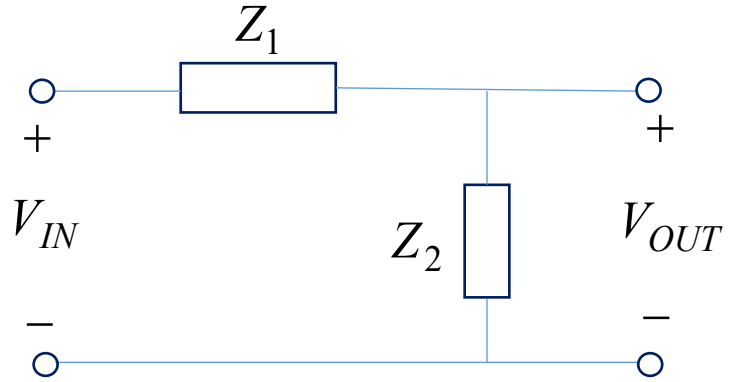
- Examples of Notch filter amplitude responses



- Are used to remove unwanted frequency from the signal with little effect on other frequencies

Low-pass Filter





$$H(s) = \frac{V_{OUT}}{V_{IN}} = \frac{\frac{1}{LC}}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$For\ R = 1\Omega,\ L = 1H;\ C = 1F$$

$$H(s) = \frac{1}{s^2 + s + 1}$$

Low-pass Filter ../3

$$H(s) = \frac{1}{s^2 + s + 1}$$

$$Let\ s = j\omega$$

$$H(j\omega) = \frac{1}{1 - \omega^2 + j\omega}$$

$$|H(j\omega)| = \frac{1}{\sqrt{(1-\omega^2)^2 + \omega^2}}$$

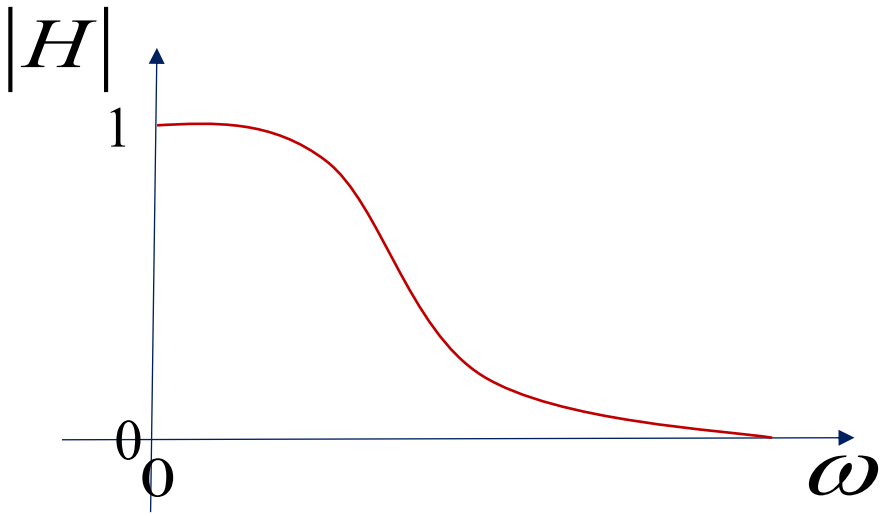
$$\angle H(j\omega) = -\tan^{-1}\left(\frac{\omega}{1-\omega^2}\right)$$

$$when\ \ \omega = 0 \Rightarrow |H(j\omega) = 1$$

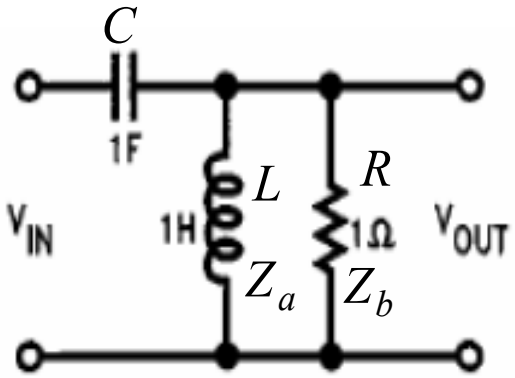
$$as\ \omega \uparrow, \quad |H(j\omega)| \downarrow$$

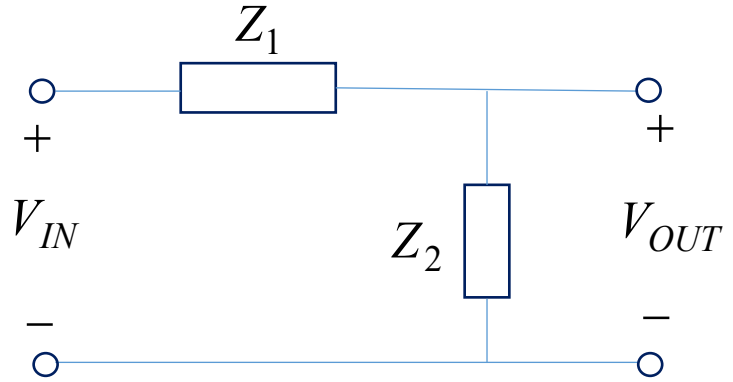
$$as\ \omega \to 0, \quad |H| \to 1$$

$$as\ \omega \to \infty, \quad |H| \to 0$$



High-pass Filter





$$H(s) = \frac{V_{OUT}}{V_{IN}} = \frac{s^2}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$For\ R = 1\Omega,\ L = 1H;\ C = 1F$$

$$H(s) = \frac{s^2}{s^2 + s + 1}$$

High-pass Filter ../3

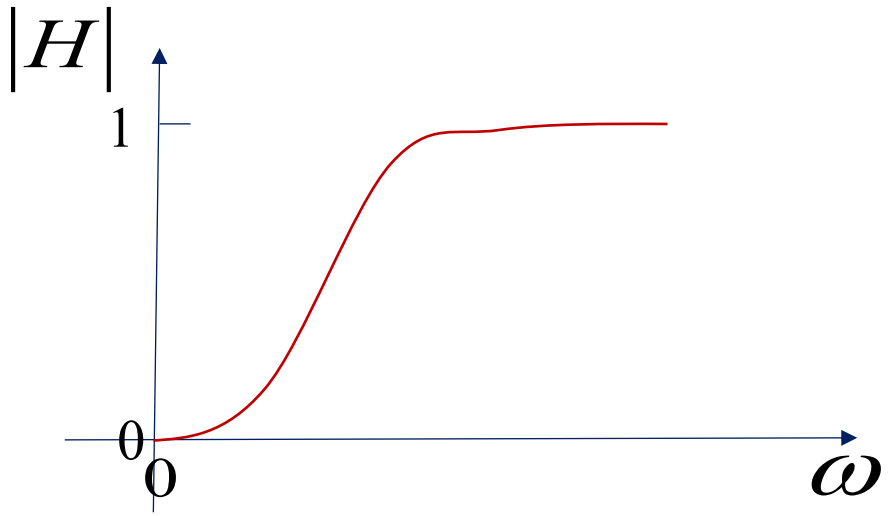
$$H(s) = \frac{s^2}{s^2 + s + 1}$$

$$Let\ s = j\omega$$

$$H(j\omega) = \frac{-\omega^2}{1 - \omega^2 + j\omega}$$

$$|H(j\omega)| = \frac{\omega^2}{\sqrt{(1-\omega^2)^2 + \omega^2}}$$

$$\angle H(j\omega) = -\tan^{-1}\left(\frac{\omega}{1-\omega^2}\right)$$



Filters

Properties that characterize 2nd order filters

a) Filter type (LPF, HPF etc)

b) The passband gain

c) The centre frequency f_c

d) The quality factor Q

Comparing Transfer Functions (TFs)

a) All TFs share the same denominator

b) Numerators are made up of terms found in the denominator
   - (HPF - s^2, B PF - s, LPF – 1, Notch - $s^2 + 1$)

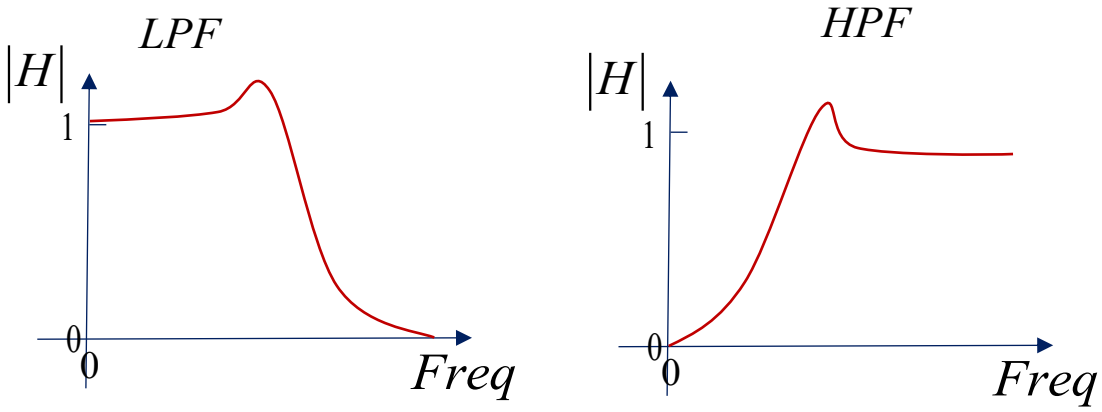
Q Factor

- Determines relative shape of amplitude response
- Can be found from denominator TF

$$D(s) = s^2 + \frac{\omega_0}{Q}s + \omega_0^2$$

- For BPF & Notch filters
  - Q relates to sharpness of amplitude response
  - As Q increases, sharpness of response increases

- LPF & HPF exhibit peaks in their response curves when Q becomes large



Q Factor ../2

- E.g. Notch filter

$$H(s) = \frac{s^2 + \frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

$$D(s) = s^2 + \frac{R}{L}s + \frac{1}{LC} = s^2 + \frac{\omega_0}{Q}s + \omega_0^2$$

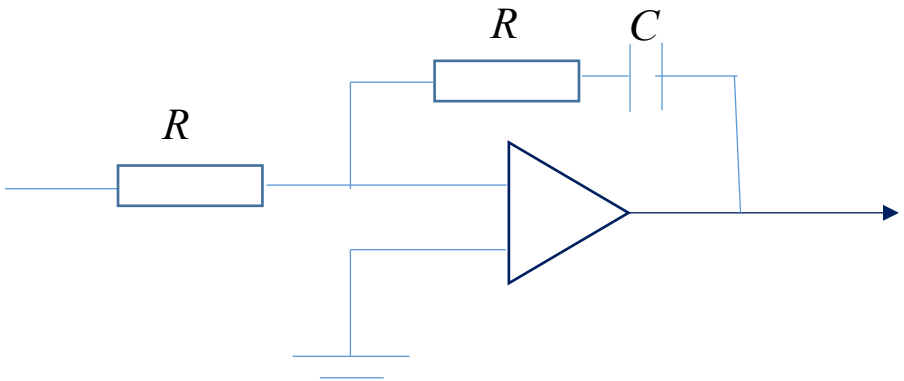
$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}; \qquad Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

- Exercise
  - Determine Q factor for BPF

Active Filters

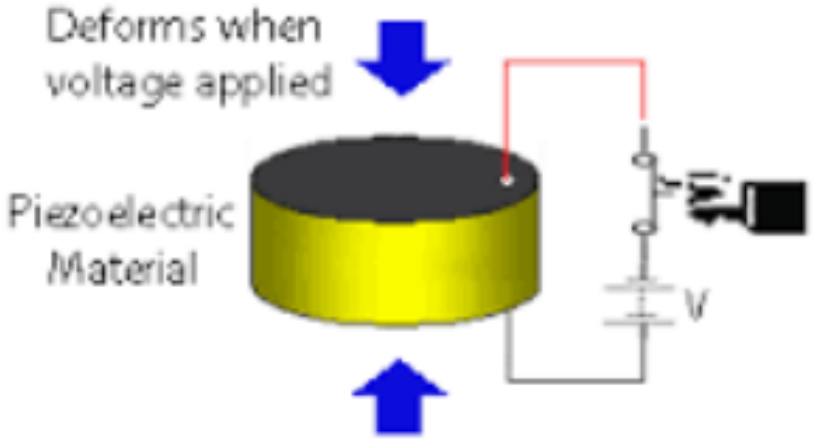
- Use amplifying elements (op amps with Rs & Cs in their feedback loops)

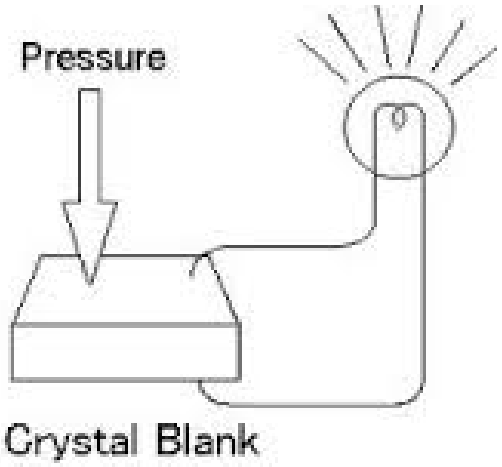
- See notes



Crystal & Ceramic Filters

- Piezo-electricity
  - Electricity produced in non conducting materials (crystals such as quartz, certain ceramics) when subjected to pressure

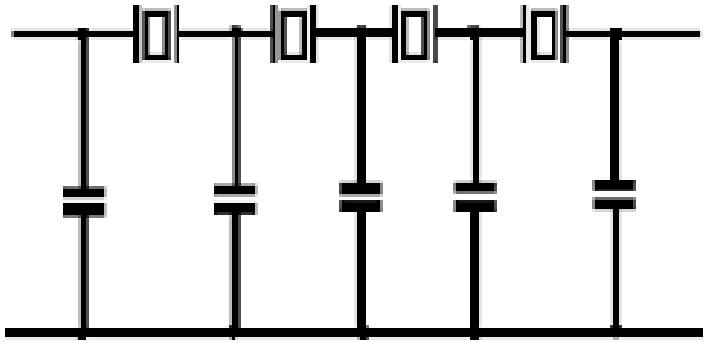


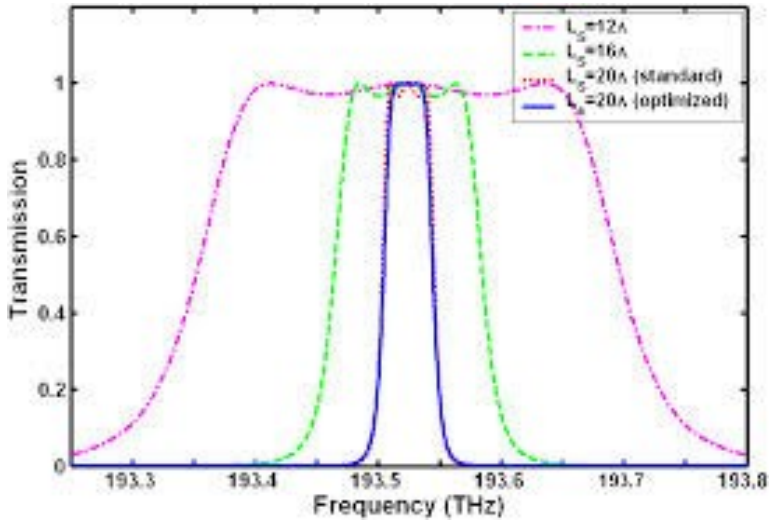


Crystal & Ceramic Filters ../2

- Properties of Quartz Crystal (quartz)
  - ➢Very high Q

- Using several crystals in a lattice
  - ➢ Bandpass filter can be made, with wider bandwidth





Ceramic Filters

- Ceramic material
  - ➢Exhibit piezoelectric effect similar to that of quartz
- Q of ceramic filters is lower than for quartz
- Bandwidth of ceramic filters is wider
- Advantage of crystal & ceramic filters?



